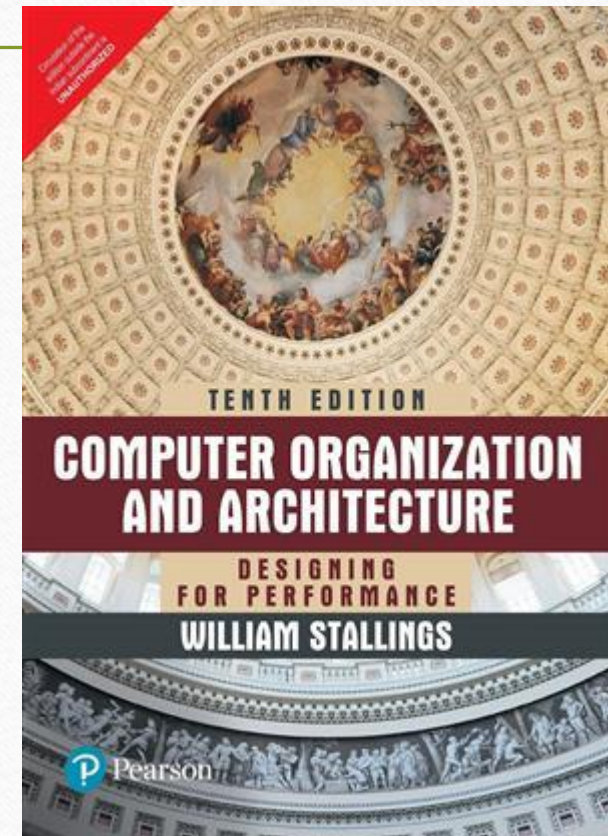


Computer Architecture and Organization

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Text Book

- Computer Organization and Architecture: Designing for Performance, 10th Ed., 2016,
 - William Stallings,
 - Prentice-Hall International, Inc.



Need of Architecture and Organization

- All students of computing should acquire some understanding of a computer system's functional components, their characteristics, their performance, and their interactions.
- Students need to understand computer architecture in order to structure a program so that it runs more efficiently on a real machine.
- In selecting a system to use, they should be able to understand the tradeoff among various components, such as CPU clock speed vs. memory size.

Computer Architecture

- Computer architecture refers to,
 - those attributes of a system visible to a programmer
 - those attributes that have a direct impact on the logical execution of a program.
 - Examples
 - Instruction set
 - The number of bits used to represent various data types (e.g., numbers, characters)
 - I/O mechanisms, and techniques for addressing memory.
- It is an architectural design issue whether a computer will have a multiply instruction.

Computer Organization

- Computer organization refers to the operational units and their interconnections that realize the architectural specifications.
- Organizational attributes include those hardware details transparent to the programmer
- Examples
 - Control signals;
 - Interfaces between the computer and peripherals;
 - The memory technology used.
- It is an organizational issue whether that instruction will be implemented by a special multiply unit or by a mechanism that makes repeated use of the add unit of the system.

Architecture and Organization

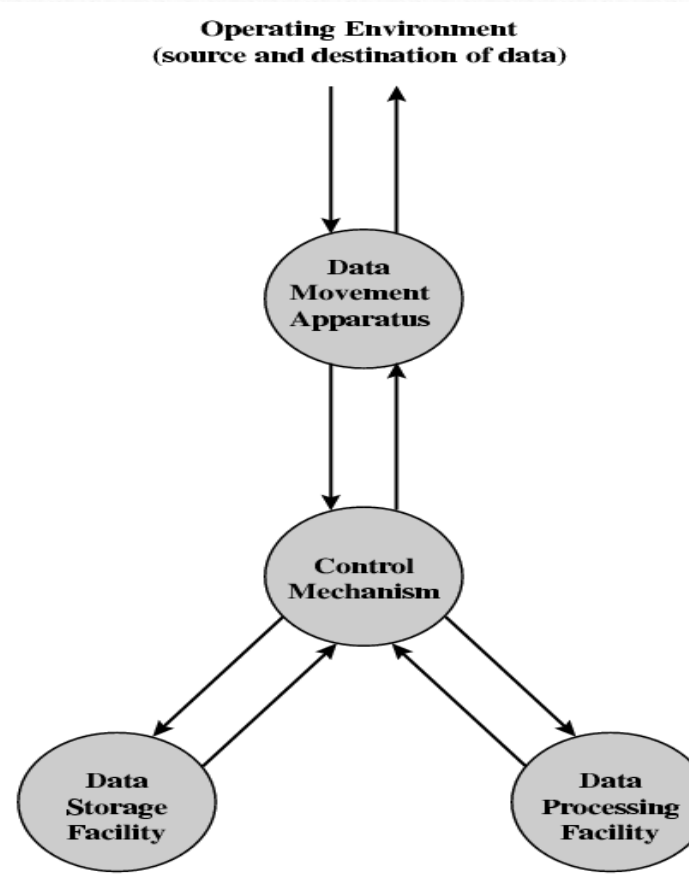
- A particular architecture may span many years and encompass a number of different computer models, its organization changing with changing technology.
- IBM System/370 architecture - This architecture was first introduced in 1970 and included a number of models.
- The customer with modest requirements could buy a cheaper, slower model and, if demand increased, later upgrade to a more expensive, faster model

Structure and Function

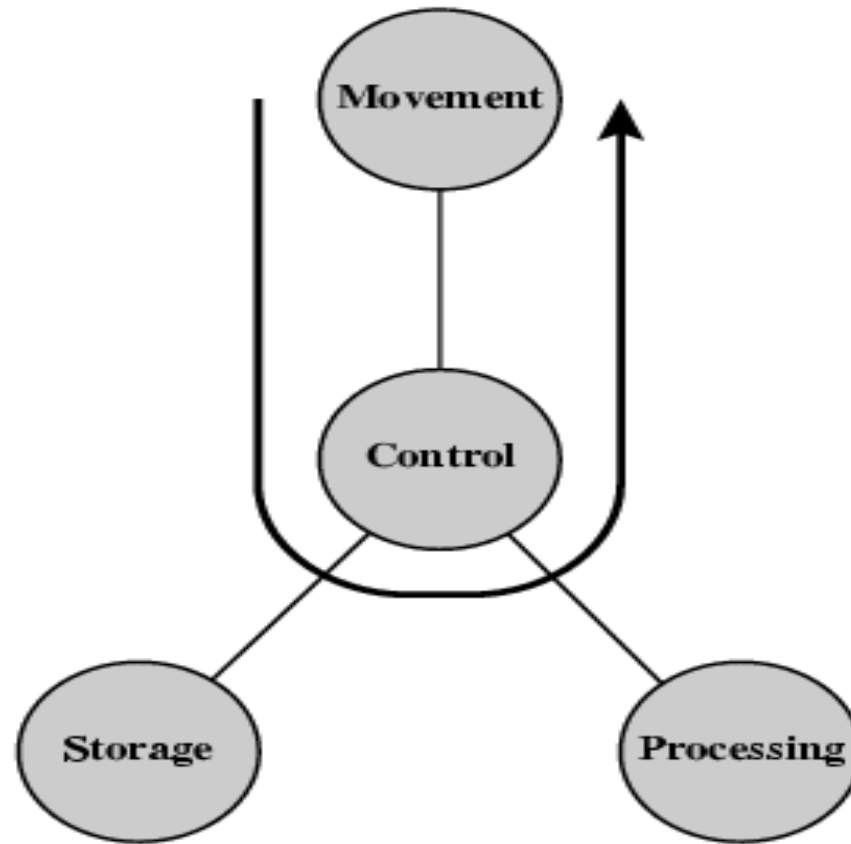
- Structure is the way in which components relate to each other
- Function is the operation of individual components as part of the structure

Functions

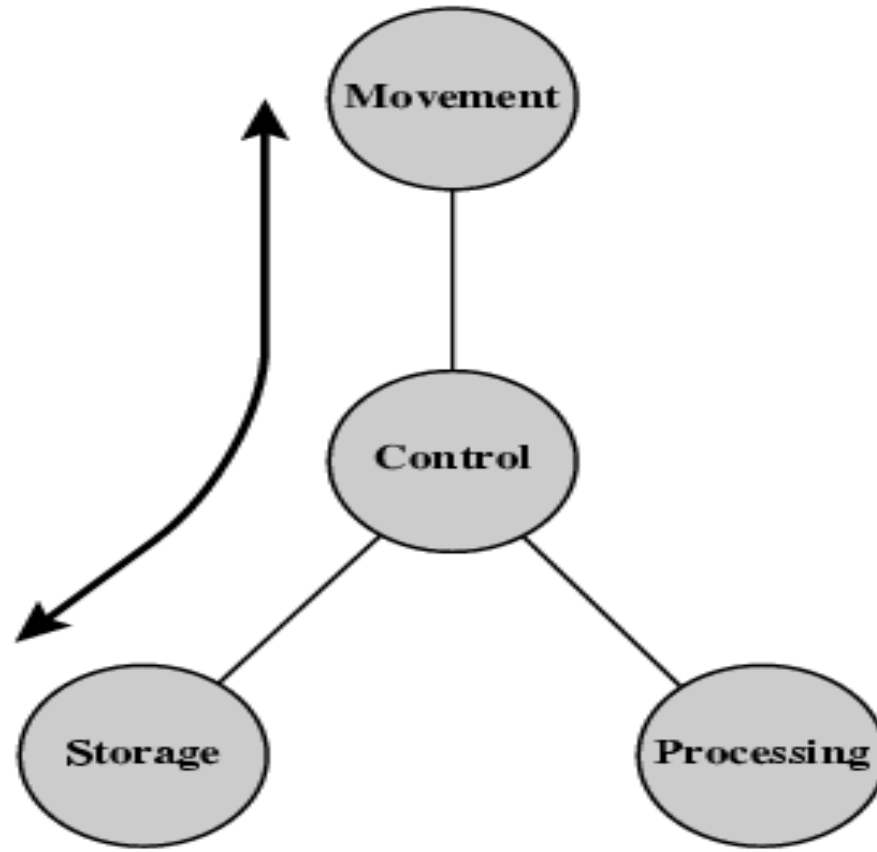
- All computer functions are:
 - Data processing
 - Data storage
 - Data movement
 - Control



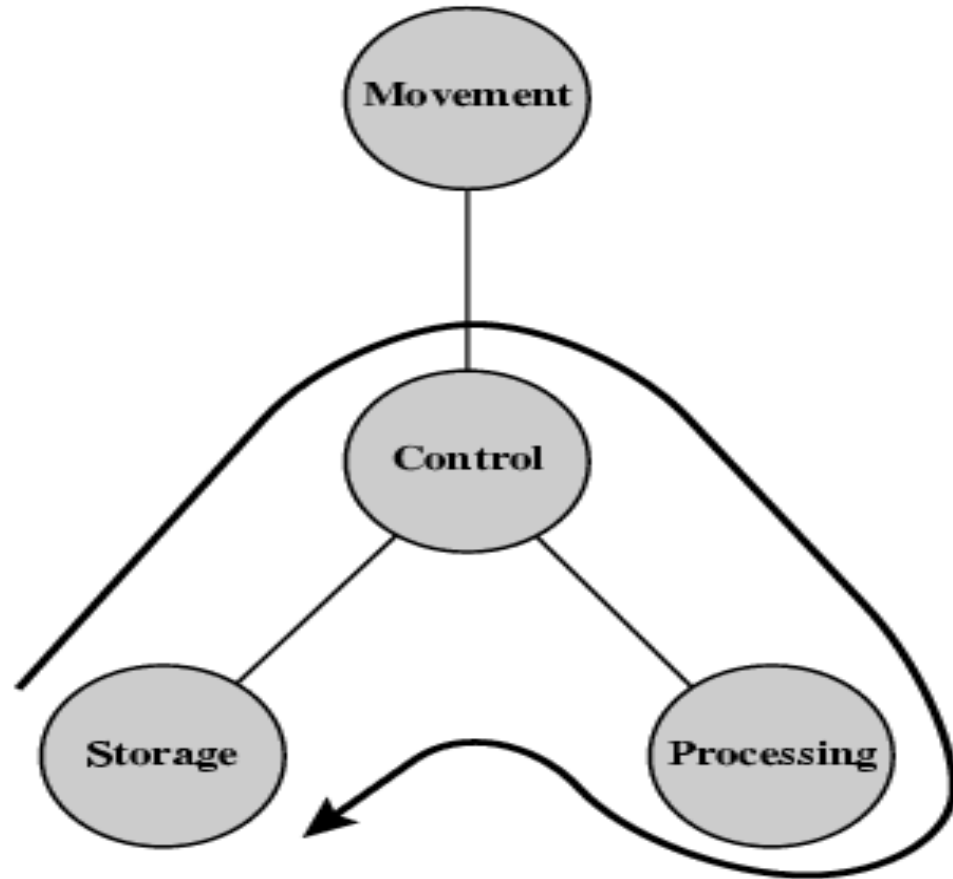
Operations (1) Data movement



Operations (2) Storage

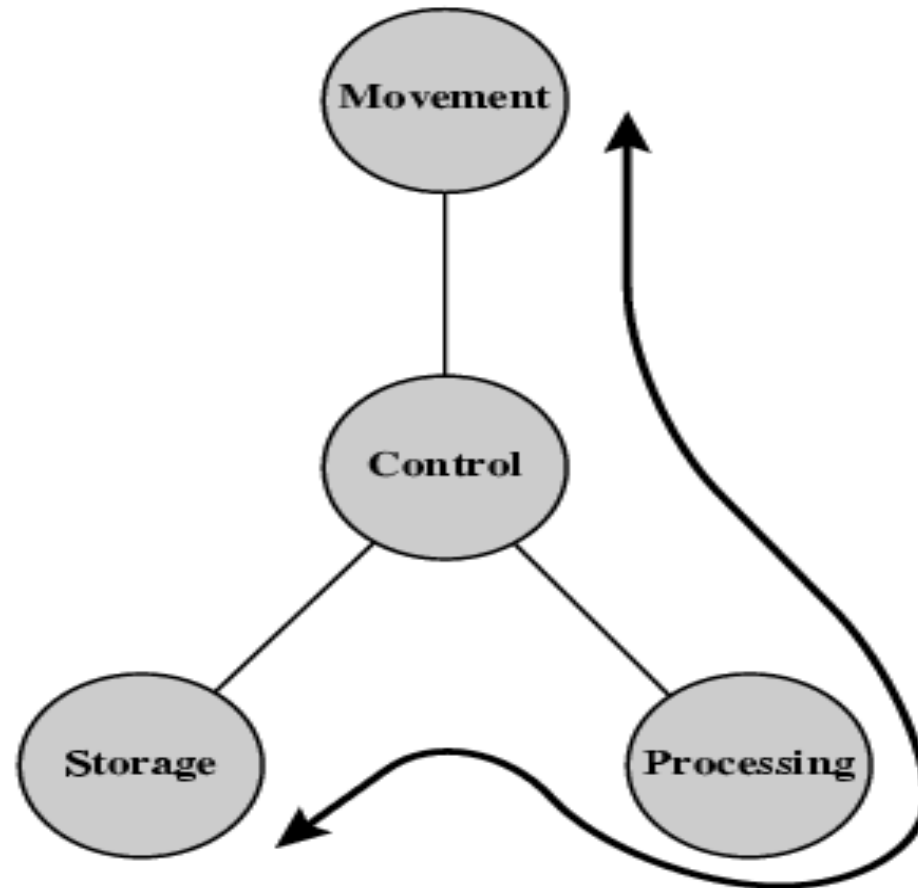


Operation (3) Processing from/to storage



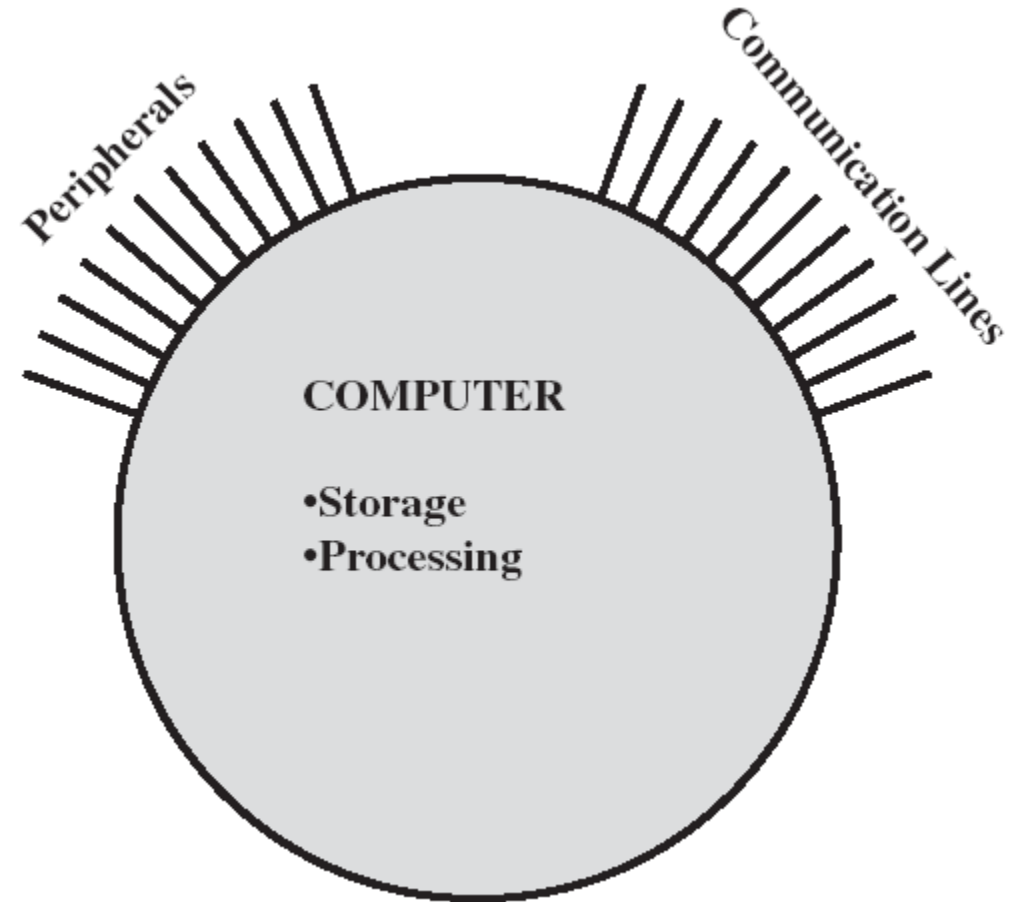
Operation (4)

Processing from storage to I/O

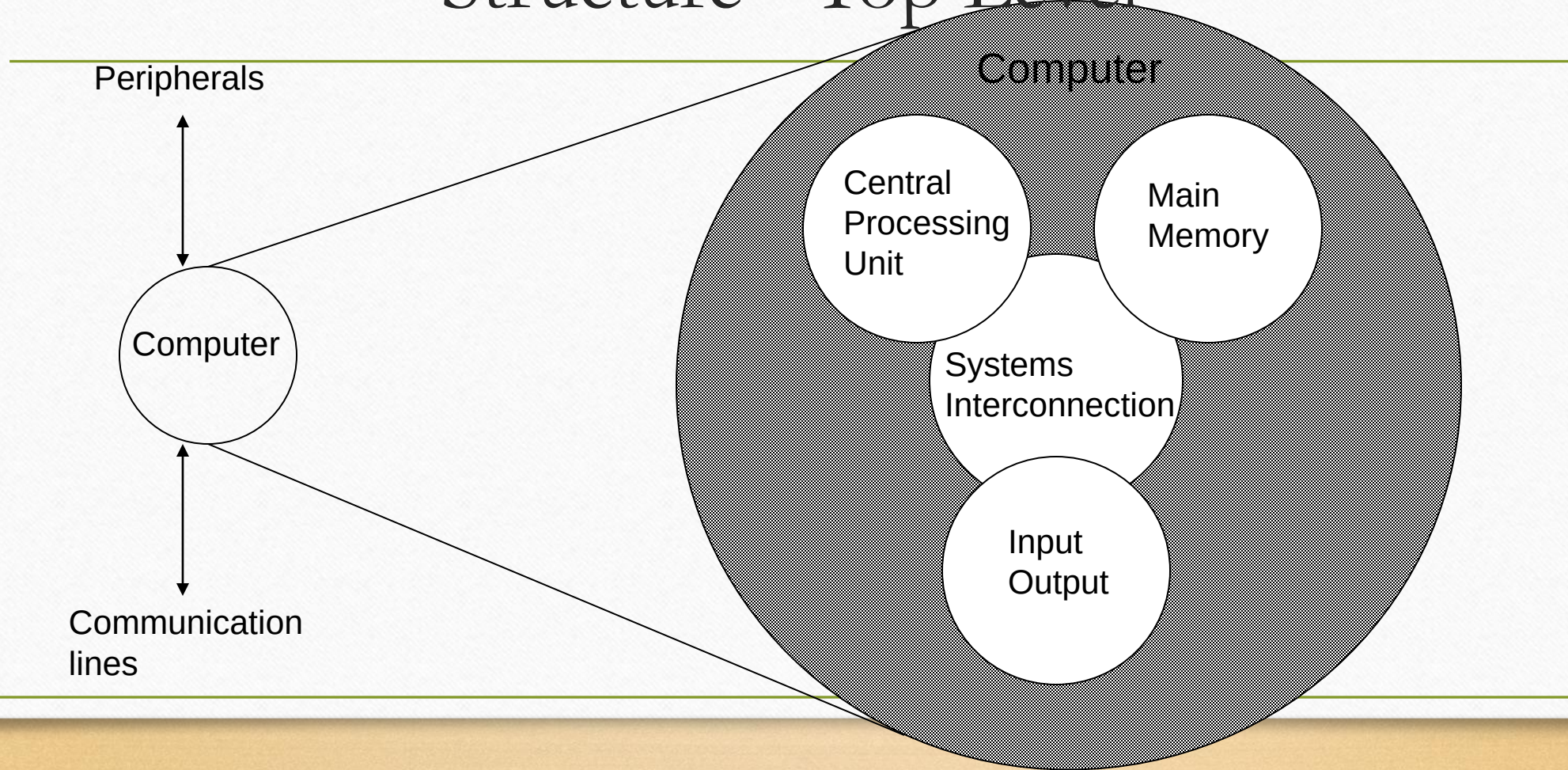


Structure

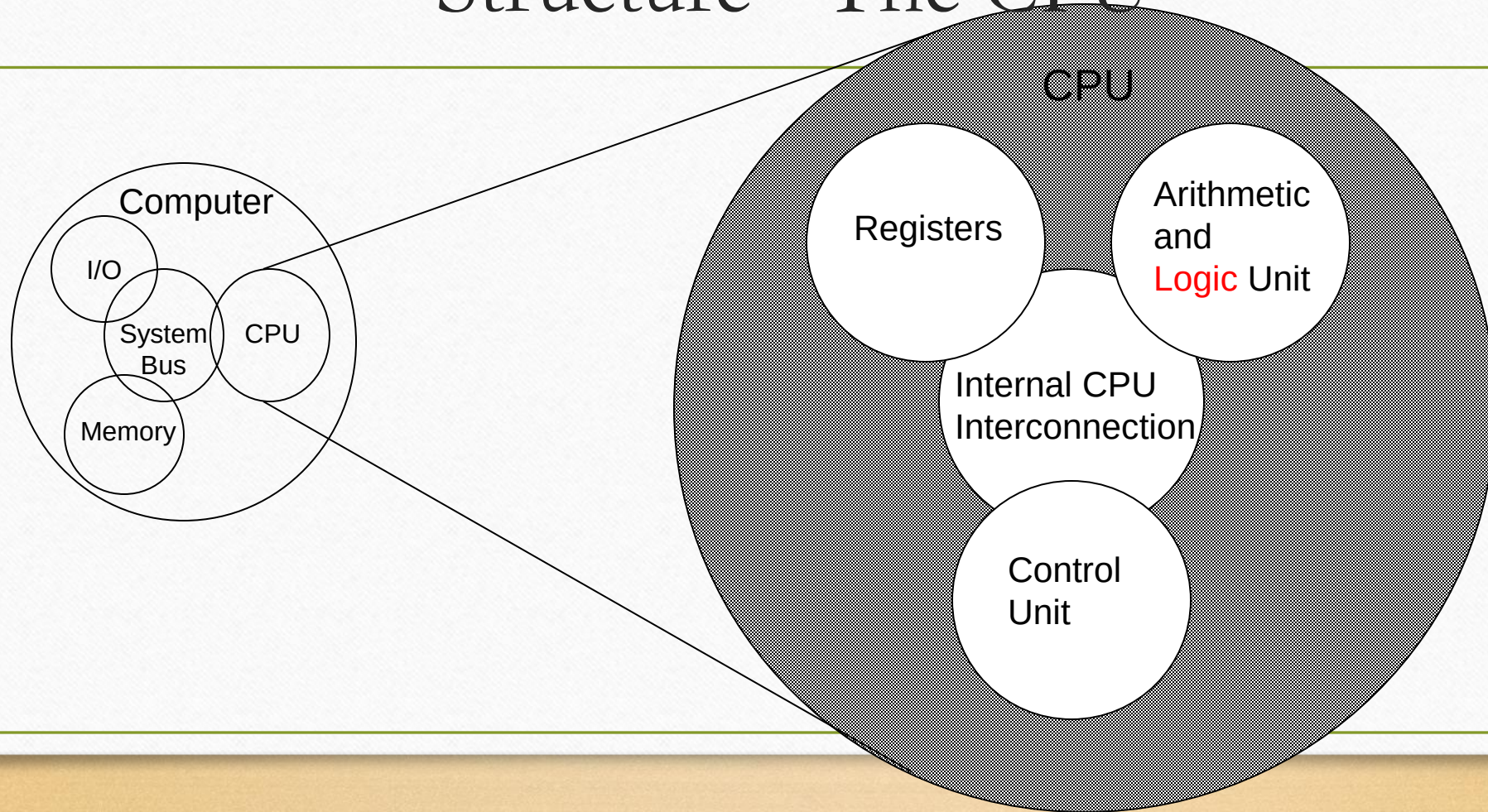
- The Computer
 - CPU
 - Controls the operation of the computer and performs its data processing functions.
 - Main memory
 - Stores data
 - I/O
 - Moves data between the computer and its external environment
 - System interconnection
 - Provides for communication among CPU, main memory, and I/O



Structure - Top Level



Structure - The CPU



Structure - The Control Unit

